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Untitled3.ipynb
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[1] import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split, cross_val_score, learning_curve, validation_curve, GridSearchCV
from sklearn.tree import DecisionTreeRegressor
from sklearn.metrics import mean_squared_error, r2_score

data = fetch_california_housing()
X = pd.DataFrame(data.data, columns=data.feature_names)
y = pd.Series(data.target)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
model = DecisionTreeRegressor(random_state=42)
model.fit(X_train, y_train)

train_pred = model.predict(X_train)
test_pred = model.predict(X_test)

train_rmse = np.sqrt(mean_squared_error(y_train, train_pred))
test_rmse = np.sqrt(mean_squared_error(y_test, test_pred))
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[1] plt.figure()
plt.bar(['Train RMSE', 'Test RMSE'], [train_rmse, test_rmse])
plt.title('Overfitting Detection')
plt.ylabel('RMSE')
plt.show()

cv_scores = cross_val_score(
    model, X_train, y_train, cv=5, scoring='neg_root_mean_squared_error'
)
plt.figure()
plt.plot(range(1,6), -cv_scores, marker='o')
plt.title('Cross Validation RMSE')
plt.xlabel('Fold')
plt.ylabel('RMSE')
plt.show()

param_range = [1, 2, 5, 10, 20, 50, 100]
train_scores, test_scores = validation_curve(
    DecisionTreeRegressor(random_state=42),
    X_train,
    y_train,
    param_name='max_depth',
    param_range=param_range,
    cv=5,
    scoring='neg_root_mean_squared_error'
)
```







